

Basics of Problem Solving

Flowchart

1. A flowchart is a graphical representation of a process, system or algorithm.
2. It uses symbols and arrows to show the sequence of steps needed to complete a task.
3. Flow charts are useful in visualising how to solve a problem and make the process easier to follow



Terminal



Input / Output



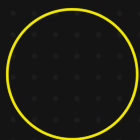
Process Box



Decision

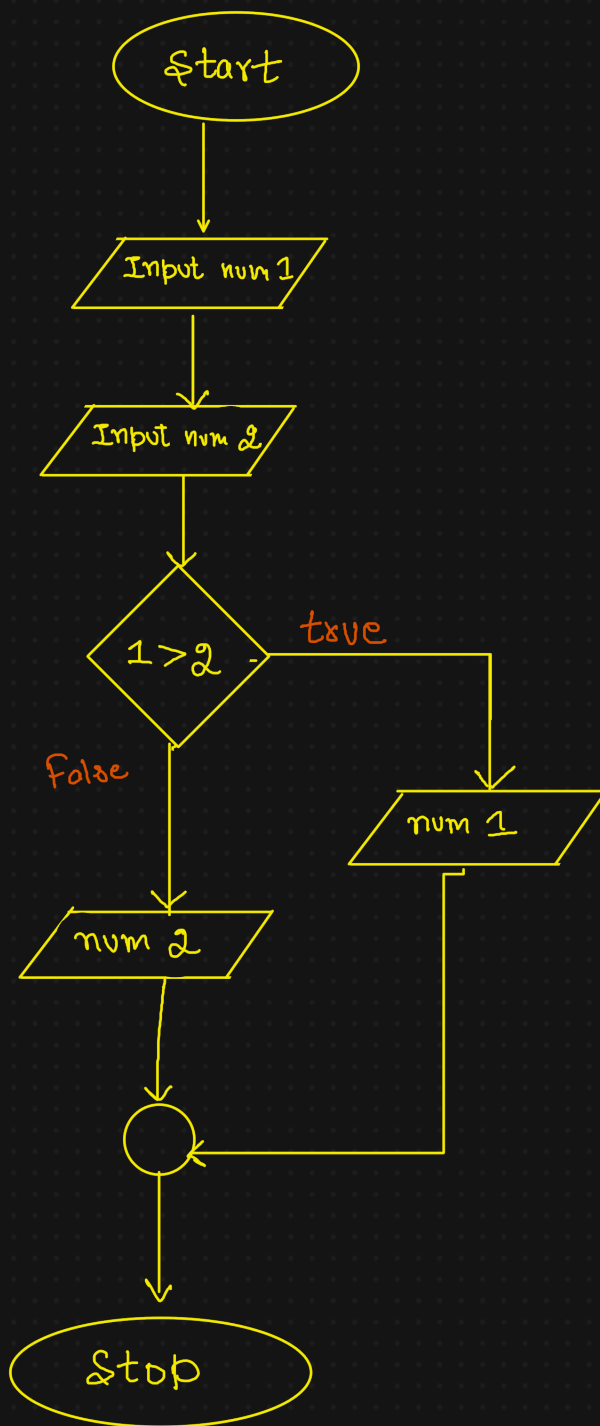


Arrow



Connector

Flowchart to display maximum of 2 numbers



Flowchart to add 2 numbers

Pseudocode

1. A **Pseudocode** is a simple way of **describing an algorithm in plain language** or a mix of code like and natural language instructions.
2. It doesn't follow any specific programming language syntax, making it easier to understand, even for those without programming knowledge
3. Pseudo code is often **used as a step before writing actual code**.

Pseudocode to find sum of 2 numbers

```
""  
begin  
input num1  
input num2  
  
sum=num1+num2  
  
print sum  
  
end  
""
```

Process to Solve a Problem

1. Analyze your Problem. Input , Output , constraint
2. Break down problem into smaller subparts.
3. Remember/enlist the concept.
4. Take 2-3 examples. Clear out confusion/gap
5. Write a pseudocode on a paper. for ($i \rightarrow n$)
6. Dry run it once.
7. Write down your solution.
8. Look out for edge cases / boundary condition. *